Data dictionary — supplementary tables and deposited derived data

Manuscript: *Evolutionary constraint reveals lineage-specific antifungal target prioritization and host-selectivity trade-offs across fungal pathogens* (mSystems submission). Written 26:10:12. Requested by the advisor's revision brief, item 6 (reproducibility package).

Supplementary tables are numbered S1 onward without gaps. Tables S1, S5, S9, S12 to S17 and S19 are supplied as separate data files; S2 to S4, S6 to S8, S10, S11 and S18 are tabulated inside Text S1. Every workbook opens with an `_INDEX` sheet mapping each sheet to the source file it was built from.

Each separate data file is one .xlsx workbook, `Table_S<n>.xlsx`, with one worksheet per source file. Worksheet names are the source file names with the `rev_b<n>_` prefix dropped and truncated to the 31-character Excel limit; the source files are named for each table below and are the canonical CSV outputs in `output/table/`. The workbooks are written by `analysis_code/build_supplemental_data_files.py`.

Separate data files

Table S1 — assembly inventory and quality control

Source: `output/table/assembly_coverage_filter.csv` (1,097 rows, one per NCBI assembly).

Column	Meaning
species	*Aspergillus* species as recorded by NCBI
accession	GenBank/RefSeq assembly accession
strain	submitter strain designation
level	assembly level (contig / scaffold / chromosome / complete)
tech, method	sequencing technology and assembly method as deposited
coverage_raw, coverage_x	submitter-reported coverage, and coverage used for filtering
total_len_Mb, expected_Mb	assembly length and the species-expected genome size
contig_n50, scaffold_n50, n_contigs, n_scaffolds, gc	standard assembly metrics
annotated	whether the record carries a deposited annotation
biosample, bioproject, date, submitter	provenance
atypical, refseq_category	NCBI flags used in the pass/fail decision
isolation_source, host, geo	metadata where deposited

Table S5 — per-gene constraint, both genera, and the ranked candidate list

Sources: `analysis/target_discovery/constraint_per_gene_dedup.tsv` (9,823 *A. fumigatus* genes), `analysis/candida/constraint/ctrop_constraint_per_gene.tsv` (6,254 *C. tropicalis* genes), `output/table/ms_table7_crossgenus_candidates.csv` (391 candidates with a one-to-one orthologue).

A. fumigatus columns:

Column	Meaning
protein_id, locus_tag	RefSeq protein accession and AFUA locus tag
contig, strand, n_codons, protein_len	coding-sequence coordinates and length
syn_sites, nonsyn_sites	counted synonymous and non-synonymous sites
syn_seg, nonsyn_seg, stop_seg	segregating synonymous, non-synonymous and nonsense sites
pi_syn, pi_nonsyn	nucleotide diversity at synonymous and non-synonymous sites
piN_piS	allele-frequency-weighted ratio, unshrunk
piNpiS_shrunk	**the *A. fumigatus* constraint estimate reported throughout** (lower = more constrained). Not stored in this sheet: it is recomputed as `((nonsyn_seg + 0.5)/nonsyn_sites) / ((syn_seg + 0.5)/syn_sites)`, and the computed value for every gene is in the `canonical_filter_audit` sheet
pN_pS	unweighted proportion-based ratio, for comparison
mean_cov_genomes, min_cov_genomes	callability: genomes contributing to the estimate
product	NCBI product description

C. tropicalis columns differ as follows: `piNpiS_dedup` — the raw πN/πS ratio on the 151 lineage representatives, **the primary *C. tropicalis* estimator and the value reported in every table and figure**; `piNpiS` — the same raw ratio on all 424 isolates; `piNpiS_shrunk` and `piNpiS_shrunk_dedup` — the genome-mean shrinkage estimator on all isolates and on lineage representatives, reported only as sensitivity analyses; and `n_iso_callable` (isolates contributing).

Table S9 — *C. albicans* essentiality calls and the cross-genus orthology map

Sources: `analysis/crossgenus/calb_essentiality.tsv.gz` (25,536 gene-call records), `analysis/crossgenus/ortho_map.tsv` (9,823 rows), `analysis/crossgenus/calb_idmap.tsv`.

Column	Meaning
gene_id, gene_name	*C. albicans* systematic and common gene identifiers
call	ESSENTIAL / NON-ESSENTIAL / other, as reported by the source
source	the published dataset or curation the call came from
evidence	the original evidence string, retained verbatim

Orthology map columns: `af_protein`, `af_locus` (*A. fumigatus*), `ct_protein`, `ct_locus`, `ct_pident`, `ct_qcov` (*C. tropicalis* reciprocal best hit with identity and query coverage), `calb_acc`, `calb_pident`, `calb_qcov` (*C. albicans* equivalent). Empty fields mean no reciprocal best hit passed the thresholds stated in Materials and Methods.

Tables inside Text S1

Each is self-describing in its legend. S2 constraint-measurement validation; S3 known and clinical-stage antifungal targets run back through the funnel; S4 the 18 Tier-1 candidates (columns: rank, locus_tag, product, piNpiS — which is the shrunk *A. fumigatus* estimator `piNpiS_shrunk`, constraint_pctile, syn_seg, nonsyn_seg, plddt, pocket, pdb, pdb_identity, ligand, human_homolog_sensitive); S6 ERG11/cyp51A residue mapping; S7 constraint versus experimental essentiality by source; S8 the cross-genus intersection annotated with essentiality; S10 landmark genes in both genera; S11 the full cross-genus intersection.

Tables added in the 26:10:12 revision

All except Table S18 are supplied as separate data files. The generating script is named for each; every script is in `analysis_code/` and writes to `output/table/`.

Table S12 — essentiality-benchmark definitions and per-model performance

Script: `rev_b1_essentiality_benchmark.R`. Files: `rev_b1_outcome_definitions.csv`, `rev_b1_performance.csv`.

Column	Meaning
source	the published *C. albicans* essentiality dataset
rule	`strict` (within-source contradictions excluded) or the parallel any-essential rule
species	genus in which constraint was measured (`af` or `ct`)

Column	Meaning
model	M1 constraint, M2 similarity, M3 covariates, M4 integrated, M1s shrunk-estimator sensitivity, B_random and B_prevalence baselines
n, n_ess, prevalence	evaluable genes, essential genes and class prevalence
n_families	gene families spanned, the unit of cross-validation and resampling
roc_auc, roc_lo, roc_hi	ROC-AUC with 95% bootstrap confidence interval
pr_auc, pr_lo, pr_hi	average precision with 95% bootstrap confidence interval

Table S13 — paired model differences and gene-family blocks

Script: `rev_b1_essentiality_benchmark.R`. Files: `rev_b1_paired_deltas.csv`, `rev_b1_gene_family_block`
Delta columns are `d_roc` and `d_pr` with `_lo`/`_hi` intervals, `_excl0` flags and a two-sided bootstrap *P* value, all computed on identical family-blocked resamples. The block file maps every *A. fumigatus* protein to its family.

Table S14 — ranking robustness and the matched cross-genus comparison

Scripts: `rev_b2_asp_recompute.py`, `rev_b2_ctrop_recompute.py`, `rev_b2_analyse.R`, `rev_b2_crossgenus.R`. Files: `rev_b2_settings_metrics.csv` (one row per parameter setting), `rev_b2_resample_metrics.csv` (one row per lineage draw), `rev_b2_candidate_percentiles.csv`, `rev_b2_tier1_retention.csv`, `rev_b2_tier1_eligibility_by_setting.csv`, `rev_b2_funnel_counts.csv`, `rev_b2_reliability.csv`, `rev_b2_crossgenus_matched.csv`, `rev_b2_crossgenus_depthmatch*.csv`, `rev_b2_crossgenus_disattenuated.csv`, `rev_b2_setlevel_jaccard.csv` and `rev_b2_top_of_list_diagnost`
Key columns are `rho_genomewide`, `tau_genomewide`, `rho_shortlist`, `jaccard_top20/50/100`, `n_eligible` and `eligible_freq`.

Table S15 — pathway and host-homology enrichment

Script: `rev_b3_pathway_enrichment.R`. Files: `rev_b3_enrichment_all_tests.csv` (every tested term, including nulls), `rev_b3_significant_after_FDR.csv`, `rev_b3_test_family_summary.csv`, `rev_b3_universe_definition.csv`, `rev_b3_gene_sets.csv`, `rev_b3_method_agreement.csv`, `rev_b3_permutation_matching_quality.csv`, `rev_b3_negative_control_calibration.csv`, `rev_b3_host_homology_gradient.csv`, `rev_b3_host_homology_by_constraint_decile.csv`, `rev_b3_crossgenus_constraint_overlap.csv`, `rev_b3_negative_control_replicates.csv` and `rev_b3_negative_control_replicate_summary.csv`. The same set of files is present a second time with the suffix `_ctshrunk_sensitivity`, and those sheets carry the `SENS_` prefix in the workbook: they are the sensitivity arm in which the *C. tropicalis* gene sets are defined on the shrunk estimator instead of the primary raw one. Each test carries `or_fisher`, `beta`, their confidence intervals, raw and Benjamini–Hochberg-adjusted *P* values for all three methods, and the permutation null mean and standard

deviation. Annotations are cached under `analysis/enrichment/` with `PROVENANCE.md` recording the retrieval date.

Table S16 — gene-family reconstruction and Tier-1 orthology confidence

Scripts: `rev_b4_01_assemble_family.sh`, `rev_b4_02_align_trim_tree.sh` and the accompanying R analysis. Files: `rev_b4_rib1_family_membership.csv`, `rev_b4_rib1_topology_tests.csv`, `rev_b4_rib1_AU_topology_tests.csv`, `rev_b4_rib1_rooting_sensitivity.csv`, `rev_b4_rib1_synteny_*.cs` `rev_b4_rib1_classII_cassette_fusion.csv`, `rev_b4_rib1_orthology_verdict.csv` (18 numbered evidence lines, each naming its source file), `rev_b4_tier1_orthology_confidence.csv` (grade plus the risk flags and the rule that produced it) and `rev_b4_tier1_paralog_detail.csv`. Alignments, trimmed alignments, tree files and logs are under `analysis/orthogroup/`.

Table S17 — association power simulation

Script: `rev_b5_power_simulation.R`. Files: `rev_b5_power_grid.csv` (sample size by effect size by allele frequency, with Wilson intervals), `rev_b5_power_TR46_scenario.csv`, `rev_b5_TR46_sample_size_summary.csv`, `rev_b5_power_fine_n_grid.csv`, `rev_b5_sample_size_for_80pct_` `rev_b5_scan_verification.csv` (thresholds and best P values re-derived per scan), `rev_b5_empirical_argument.csv` (observed P values only, no simulation), `rev_b5_model_validation.csv` `rev_b5_null_calibration.csv`, `rev_b5_variance_calibration.csv`, `rev_b5_dispersion_rescaling.csv`, `rev_b5_cohort_expansion_check.csv`, `rev_b5_variant_clustering.csv`, `rev_b5_simulation_parameters.c` and `rev_b5_headline_numbers.csv`.

Table S16 — gene-family reconstruction and Tier-1 orthology confidence (addendum)

The workbook also carries `rev_b4_tier1_orthology_confidence_union26.csv` and `rev_b4_tier1_paralog_detail_union26.csv`: the identical audit re-run over the 26 lead candidates (the 18 Tier-1 genes plus the 8 further genes carried in main Table 1). The 18 Tier-1 grades reproduce exactly, which is the agreement check for the published values.

Table S19 — the candidate evidence table

Scripts: `rev_c0_calb_essentiality_per_locus.R`, `rev_c1_canonical_filter_audit.R`, `rev_c2_00_candidate_set_union.R`, `rev_c2_candidate_evidence_master.R`, `rev_c3_main_table1.R`, and `rev_b4_05_tier1_orthogroup_check.R` run over the 26-gene set. One row per lead candidate, 26 in all. The 26 are the union of the 18 Tier-1 candidates, which are derived from `ms_table3_tier1.csv`, and the 12 genes carried in main Table 1, which are a declared editorial selection rather than a filter output; `rev_c2_00_candidate_set_union.R` records both.

Column group	Meaning
constraint	`piNpiS` (shrunk *A. fumigatus* estimator), `constraint_pctile` (genome-wide, among evaluable genes), `callable_genomes`, `syn_seg`, `nonsyn_seg`, `segregating_sites`, `expected_variants`
robustness	`pctile_median`, `pctile_q05`, `pctile_q95`, `rank_median`, `rank_q05`, `rank_q95`, `eligibility_retention` — the fraction of 100 lineage resamples in which the gene stays eligible
human homology	`homology_search_default` and `homology_search_sensitive` (best human hit under each search sensitivity), `homology_sens_pident`, `homology_sens_qcov`, `homology_sens_evalue`, `homology_threshold`, `homology_decision`
paralogy and orthology	`n_af_inparalogs`, `best_inparalog_acc`, `best_inparalog_pident`, `orthology_confidence` (HIGH / MEDIUM / LOW / UNRESOLVED), `n_risk_flags`, `risk_flags`, `domain_mismatch_class`, `ct_orthologue`, `ct_reciprocal`, `calb_orthologue`, `calb_reciprocal`, and `confidence_rule`, which states the grading rule in full
structure	`mean_plddt`, `n_pockets`, `best_druggability`, `best_pocket_volume`, `best_pdb_id`, `best_pdb_identity`, `top_ligand`, `structure_caveat`. Empty for genes outside the top 200, which were never assessed
external validation	`calb_acc`, `calb_pident`, `calb_qcov`, `n_src_essential` of `n_src_reporting`, `sources_essential`, `external_validation_caveat`. These describe the *C. albicans* orthologue and are not validation of the *A. fumigatus* gene
funnel status	`eligible_published`, `eligible_corrected`, `removal_reason`

Column group	Meaning
triage	`triage` (eligible / conditional / exploratory) and `triage_rationale`. These are a different decision from the funnel columns above: of the 26 genes, 23 have `eligible_published` true, 17 have `eligible_corrected` true, 6 have `homology_decision` = cut, and 7 are eligible. The cascade is evaluated in order and the first match wins, so the classes do not overlap — (i) exploratory if a human homologue is detectable at high sensitivity; (ii) conditional if the gene fails a non-selectivity criterion for a documented reason; (iii) exploratory if it fails any other corrected-funnel criterion; (iv) conditional if it passes the corrected funnel but has orthology confidence LOW or UNRESOLVED or an in-paralogue; (v) eligible otherwise. Coded in `rev_c2_candidate_evidence_master.R`

Supporting sheets: `main_table1` (main Table 1 as published, generated from this table), `tier1_reevaluation` (all 18 Tier-1 genes re-evaluated against the corrected funnel), `tier1_orthology_conf~ce_union26` and `tier1_paralog_detail_union26` (the orthology-confidence grades and in-paralogue detail for all 26 candidates, the same two files that Table S16 carries), `calb_essentiality_per_af_locus` (per-locus essentiality for every *A. fumigatus* gene with a *C. albicans* orthologue, under both the strict and any-essential collapse rules) and `essentiality_crosscheck` (agreement with the published `ms_table10_intersection_essentiality.csv`; the seven disagreements are all the GRACEv1_OMeara2015 source, which the orf19 identifier crosswalk recovers and the published table missed).

Table S18 — recommended first experiments

Tabulated in Text S1 §S2.5; no separate data file.